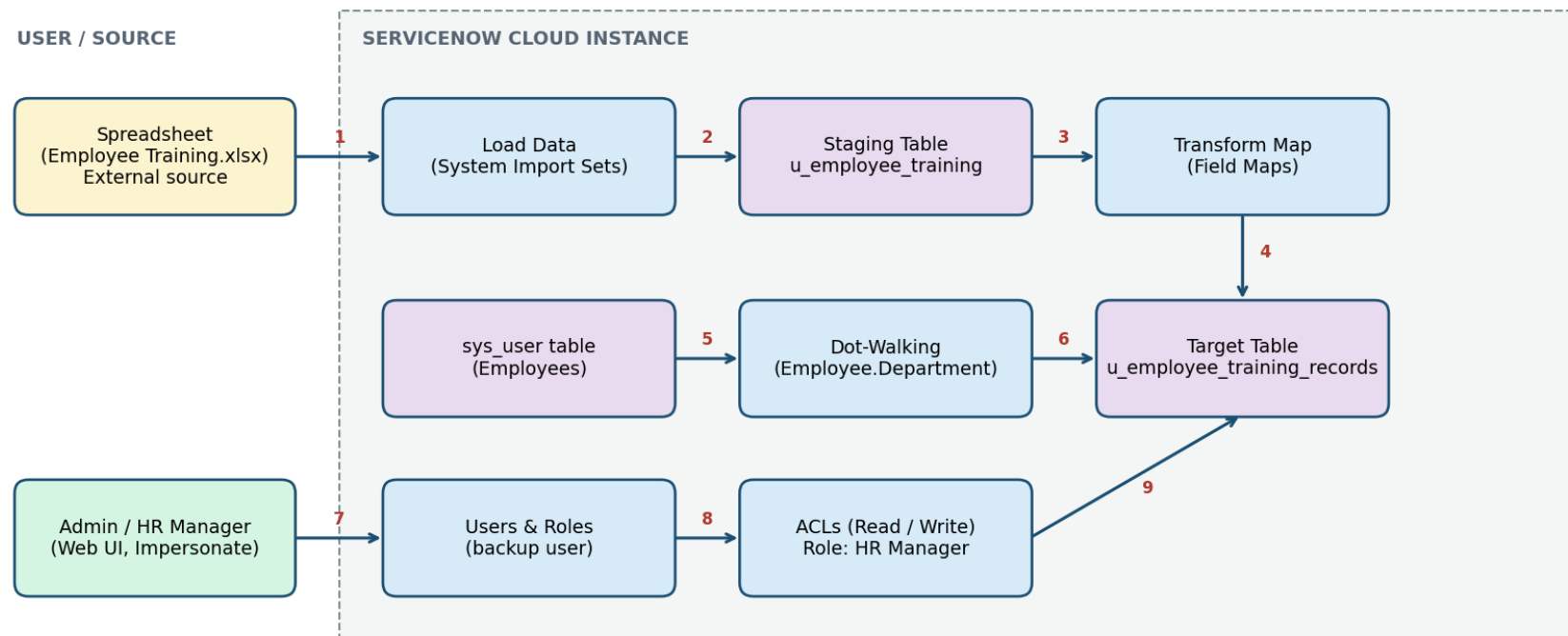


Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	6ab4d8aca238dc999eb5c33f
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

Architecture of importing spreadsheet data into ServiceNow using Import Sets and Transform Maps, linking records to employees through dot-walking, and securing them with role-based ACLs.



Flow: (1) Spreadsheet uploaded via Load Data → (2) rows stored in staging table → (3) Transform Map applies field maps → (4) data written to target table → (5)(6) Employee.Department pulled from sys_user by dot-walking → (7)(8)(9) user with HR Manager role passes ACL check to read/write records.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	ServiceNow web UI used by Admin / HR Manager to load data, create tables, forms and view records	ServiceNow Web UI (Next Experience), Form Layout / List view
2	Application Logic-1	Import of spreadsheet data into staging table and field mapping to target table	ServiceNow Import Sets (Load Data), Transform Maps
3	Application Logic-2	Dot-walking to show Employee's Department inside the training record	ServiceNow Form Layout Configuration (Employee.Department)
4	Application Logic-3	Role-based access: HR Manager role, user creation, ACL for read & write, testing by impersonation	ServiceNow ACL, Roles, Users, Impersonate User
5	Database	Custom table u_employee_training_records (Training Name, Completion Date, Status, Employee), staging table u_employee_training, reference to sys_user	ServiceNow relational database (Now Platform tables)
6	Cloud Database	Database service hosted with the ServiceNow instance	ServiceNow Cloud Instance
7	File Storage	Spreadsheet file (.xlsx) with 4 fields uploaded during Load Data	Local file upload (Excel / Google Sheets export)
8	External API-1	Not applicable – data is imported from a spreadsheet file, no third-party API used	N/A
9	External API-2	Not applicable	N/A
10	Machine Learning Model	Not applicable for this project	N/A
11	Infrastructure (Server / Cloud)	Application configured and deployed on the ServiceNow cloud instance. Local Server Configuration: not required (browser access only). Cloud Server Configuration: ServiceNow hosted instance	Cloud – ServiceNow Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework used; built with ServiceNow's low-code platform features	ServiceNow native features (Import Sets, Transform Maps, ACL)
2	Security Implementations	Role-based access control; Read and Write ACLs restricted to HR Manager role; security_admin elevation to create ACLs; access verified by impersonating users	ACL, Roles, Elevate Role, Impersonate User

S.No	Characteristics	Description	Technology
3	Scalable Architecture	Staging table separated from target table, so large spreadsheets can be loaded and transformed in repeatable batches; tables and roles reusable for more records and users	ServiceNow platform, Import Sets
4	Availability	Cloud-hosted instance accessible from any browser; availability managed by the platform	ServiceNow Cloud
5	Performance	Records transformed in bulk by Transform Map; dot-walked field avoids duplicate data entry; list view loads data directly	Transform Map, Dot-walking, List view